

Computational Modelling - Social Networks

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1 Summary

This report documents a computational agent-based model, namely ABM, that simulates the diffusion of sustainable behaviour among households in a social network. The model is implemented in Python using the Mesa framework and NetworkX library, and explores how social norms, individual psychological traits, and network structure shape together the transition towards sustainable lifestyles.

The code can be found on the Github repository in this link.

2 Introduction and Motivation

In order to design policies to tackle the problem of sustainability, to understand how sustainable practices spread through society becomes a central challenge. Other models treat populations as homogeneous, like Bansal's [1], which ignores the role of social network topology and individual-level heterogeneity. However, ABM offers a more realistic alternative, capturing the uprising overall outcomes that come from micro-level interactions.

This model, based on the example of a virus in the Github repository [5] made by Mesa focuses on three interacting forces:

- **Social influence:** Neighbors who have already adopted sustainable behavior make pressure on others.
- **Individual heterogeneity:** Each household has different openness, resistance, and inertia.
- **Network structure:** A scale-free (Barabási-Albert) topology replicates the hub-and-spoke structure of real social networks. This will be furthered discussed later in Section 6.1.1.

3 Technical Architecture

The notebook imports the following Python packages:

- **Numpy**: Random seeding, array shuffling for agent activation order.
- **NetworkX**: graph construction and neighbour traversal.
- **Mesa**: ABM framework, with agents, grid, and data collection.
- **Matplotlib**: Visualization of adoption dynamics.
- **enum**: Type-safe state representation for agents.

4 Glossary

- **ABM**. Term that means Agent-Based Model, a simulation where individual agents follow local rules, producing emergent system level behaviour.
- **Mesa** [9]. A Python library for agent-based modelling, providing Model, Agent, Grid, and DataCollector classes.
- **NetworkX libr** [4]. NetworkX is a Python package for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks.

4.1 Agent Behaviour

4.1.1 The Enum

Agent behaviour is governed by a discrete model of three states defined using Python's *Enum*:

```
class State(Enum):
    NON_SUSTAINABLE = 0 # baseline; not yet transitioning
    TRANSITIONING = 1   # in the process of adopting
    SUSTAINABLE = 2     # fully adopted
```

Using **Enum** makes state comparisons more readable. The model thus follows a one-way pipeline:

agents can never revert from SUSTAINABLE to TRANSITIONING,

which reflects the empirical observation that sustained habit changes are rarely reversed.

5 Agent Design. HouseholdAgent.

5.1 Behavioural Traits

Each household agent is initialised with three stochastic behavioural parameters drawn from uniform distributions:

- **susceptibility:** Represents the openness to change, with distribution set to $U[0.3, 1.0]$. The higher it is, the more likely to begin transitioning. The minimum of 0.3 ensures no agent is completely closed off.
- **entitlement:** Represents the resistance or the perceived exemption that the agent feels, with distribution $U[0.0, 1.0]$. The higher it is, the more dampen the adoption probability becomes. It captures how privileged households resist behavioural demands.
- **habit:** Represents the inertia or status quo bias, with distribution set to $U[0.0, 1.0]$. High values make the system sticky since both transition and stabilization become less likely.

5.2 Transition Mechanics

5.2.1 From NON_SUSTAINABLE to TRANSITIONING

At each time step, a non-sustainable agent surveys its immediate network neighbours and computes the social norm signal given by the formula:

$$\text{social_norm} = \frac{\text{sustainable_neighbours}}{\text{total_neighbours}}$$

This fraction is followed by accompanying it with the agent’s personal traits and the global social pressure represented by the parameter β to yield the **adoption probability**:

$$\text{prob} = \beta \times \text{social_norm} \times \text{susceptibility} \times (1 - \text{entitlement}) \times (1 - \text{habit})$$

The four multiplicative factors of our adoption probability model act as a gate:

- β : Represents the institutional strength of social norms, like awareness campaigns or visible green infrastructure.
- **social_norm**: represents local peer influence. If a household’s neighbours are mostly sustainable, the pressure will increase.
- $(1 - \text{entitlement})$: the more entitled an agent is, the lower the effective adoption pressure.
- $(1 - \text{habit})$: a strong habit behaviour suppresses the possibility of beginning a transition.

A uniform random draw determines whether the agent actually transitions or not. This stochastic element captures in an efficient way how unpredictable individual behavioural change is.

5.2.2 From TRANSITIONING to SUSTAINABLE

The agents that have started a transition then try to stabilize each step via

$$\text{stabilize_prob} = \gamma \times (1 - \text{habit}).$$

The parameter γ is the global stabilization rate. Very rooted habits make the stabilization process delay, which created a potential large household population that is stuck in transition when habit values are high and γ is low. This case is seen in real life. For instance, when households install solar panels but fail to sustain wider lifestyle changes.

6 Model Design. AdoptionModel.

6.1 Network Structure

6.1.1 Network Topology: Barabási-Albert Graph

The social network is generated using the Barabási-Albert, namely BA, preferential attachment model with parameters $N = 10,000$ and $m = 3$. Note that N represents the number of agents (a tractable representative sample of Amsterdam’s approximately 400,000 households, as recorded by CBS in 2023) and m the number of edges each new node forms (each new node connect to 3 existing nodes). Then,

```
self.G = nx.barabasi_albert_graph(N,3)
```

The BA graph produces a degree distribution in which most nodes have few connections, while a small number of nodes act as highly connected hubs. This has important implications for the dynamics such as:

- A small number of highly-connected nodes (called hubs) are likely to reach SUSTAINABLE early.
- Once a hub adopts, its many neighbours experience a sharp increase in social_norm, potentially triggering an adoption that looks like a cascade.
- Peripheral nodes with very few connections are more isolated and may never experience enough peer pressure.

We have included (commented out) some other alternatives for experimenting with different social topologies such as:

- Erdős-Rényi [3]: Used for a random and egalitarian society. The expected effect is to have a slower, more uniform diffusion, without a cascade effect.
- Watts-Strogatz [4]: Used for tight communities with occasional long-range ties. The expected effect is that there will be cluster level norm formations, where social norms will dominate.

Many more networks can be found on the NetworkX webpage.

6.2 Initialization and Seeding

The model starts with $N = 10,000$ household agents placed on the network. To start the diffusion, the top 400 agents by network degree (the top 0.1% of hubs) are immediately set to SUSTAINABLE:

```
sorted_nodes = sorted(self.G.nodes(),
    key=lambda n: self.G.degree[n], reverse=True)
hub_nodes = sorted_nodes[:400]
for node in hub_nodes:
    self.agent_list[node].state = State.SUSTAINABLE
```

This seeding choice is deliberate. In the BA network, adoption probability is driven by `social_norm`, the fraction of an agent’s neighbours that are already sustainable. Seeding hub nodes means their state is immediately visible to the greatest number of agents in the network, maximising the initial social norm signal across the population. Seeding randomly selected or peripheral nodes instead would leave most agents with a `social_norm` ≈ 0 for many early steps, stalling diffusion entirely. This follows Rogers’ *Diffusion of Innovations* [6]: early adopters tend to be more socially connected opinion leaders whose behaviour is most observable to others. In a BA network, hubs are the natural manifestation of this role.

6.3 Global Parameters

As a summary, you can see a recap of all the global parameters in Table 1.

Parameter	Default Value	Interpretation
β	0.25	Strength of social norm transmission. Low β implies norms matter little, high β implies strong peer pressure accelerates adoption.
γ	0.05	Rate at which transitioning agents stabilize into fully sustainable behaviour. Low γ creates a large limbo situation.
N	10,000	Total number of household agents in the simulation (representative sample of Amsterdam’s $\approx 400,000$ households).
T	200	Number of time steps, it is reasonable to think about months when interpreting it.
<code>policy_shock_time</code>	None	Time step at which an optional policy intervention is applied. Set to an integer to activate.
<code>policy_type</code>	None	String identifier of the policy to apply. See Section 8.1 for available options.

Table 1: Global Parameters Table.

6.4 Data Collection

Mesa’s DataCollector tracks three aggregate counts per step:

- Sustainable, number of adopted agents.
- Transitioning, number of agents in the middle phase.
- Non Sustainable, number of agents yet to begin.

These are later normalized to proportions and visualized as a line chart.

6.5 Simulation Loop and Agent Activation

Each call to `model.step()` does the following:

1. Applies any scheduled policy intervention if `policy_shock_time` has been reached (see Section 8.1).
2. Collects the current state counts via DataCollector.
3. Shuffles the agent list (random asynchronous activation), preventing order bias and is more realistic than sequential activation.
4. Iterates through all agents, calling `agent.step()` for each.
5. Increments the model time counter.

7 Preliminary Results

The model runs for 200 discrete time steps with $N = 10,000$ agents, $\beta = 0.25$, and $\gamma = 0.05$.

```
model = AdoptionModel(N=10_000, beta=0.25, gamma=0.05)
for _ in range(200):
    model.step()
```

The line chart of normalized population shares shows three characteristic phases. A first early phase (steps 1 to 20) where the NON_SUSTAINABLE population dominates. The hub seed of 400 sustainable agents begins to exert social norm pressure on their many immediate neighbours, producing a faster initial signal than random seeding would. Adoption is still slow, however, because the fraction of sustainable neighbours for most peripheral agents remains very low. The following is a growth phase (steps 20 to 70) where we see that as further hubs transition, a multiplier effect occurs. That is, many neighbours simultaneously experience higher social_norm values. The TRANSITIONING population thus temporarily rises and adoption accelerates. Finally, we reach a saturation phase (steps 70 to 200) where the TRANSITIONING population remains elevated as agents are slow to fully stabilize, given that our γ is so low. The NON_SUSTAINABLE falls towards a persistent residual, as there

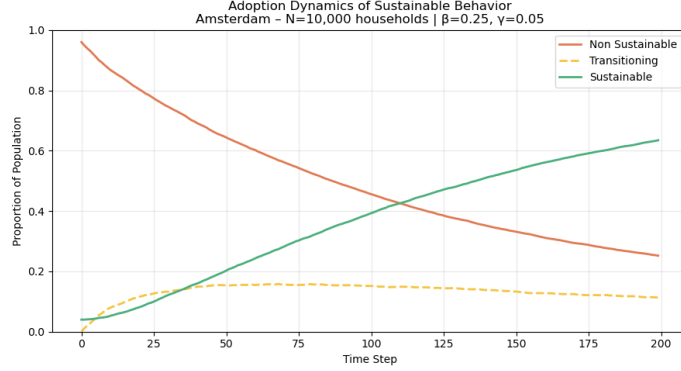


Figure 1: Line plot of model adoption dynamics of sustainable behaviour. A vertical dotted line is drawn automatically when a policy shock is active, marking the time step of the intervention.

are agents with high entitlement or habit values who are effectively immune to social pressure.

The parameter being $\beta = 0.25$ reflects a society with some norm reinforcement but no strong mandate or policy. On the other hand, $\gamma = 0.05$ reflects the difficulty of fully completing a lifestyle change. Changing one behaviour does not instantly produce a fully sustainable household. This asymmetry between β and γ creates a TRANSITIONING pool. This aligns with reality since it mirrors the empirical phenomenon of having sustainable intentions but not committing to a change all the way.

8 Possible Extensions to the Model

The code already contains several commented-out block that are ready to use extensions for the model.

8.1 Policy Shocks

The model supports four ready-to-use policy interventions, each grounded in empirical literature. A policy is activated by passing `policy_shock_time` and `policy_type` when creating the model, for example:

```
model = AdoptionModel(N=10_000, beta=0.25, gamma=0.05,
                      policy_shock_time=50,
                      policy_type='carbon_tax')
```

Here are some policy shocks based on the literature about how this would affect behaviour. Note, however, that they are educated guesses based on the information given in the papers:

- **carbon_tax**: Increases β by 0.15, raising the perceived social cost of non-sustainable behaviour. Stern [8] documents how price signals increase the strength and adherence of norms.
- **awareness_campaign**: Reduces each agent's **entitlement** by 30%, simulating a city-wide information campaign that lowers psychological resistance. Based on Schultz et al. [7], which shows that descriptive norm messaging systematically reduces self-exemption beliefs, namely entitlement.
- **habit_subsidy**: Reduces each agent's **habit** by 40%, modelling instruments that lower the cost of breaking inertia, such as heat-pump subsidies. Based on Verplanken & Wood [10] on habit discontinuity and the Dutch SDE++ subsidy evaluation literature.
- **influencer_seeding**: Forces the top 1% of highest-degree nodes to SUSTAINABLE at the shock time, mimicking targeted outreach to community leaders such as people with a high number of followers in social media. Based on Centola [2], which demonstrates that seeding hubs produces disproportionately fast network-wide diffusion relative to random seeding.

When a policy shock is active, the output line chart displays a vertical dotted line at **policy_shock_time**, making it straightforward to compare pre- and post-intervention dynamics across scenarios.

8.2 Elite Resistance

```
if self.model.G.degree[self.pos] > 10:
    self.entitlement = 0.8
```

High degree nodes, namely the hubs, could be assigned high entitlement, making them resistant adopters. By doing so, we would test whether the diffusion model is robust when the most influential agents are also the most resistant, a phenomenon observed in high-income households.

8.3 More Suggested Extensions

In addition to these, we could include future extensions such as:

- **Reversibility**: Allow SUSTAINABLE agents to revert to TRANSITIONING under certain conditions such as economic shocks, creating a more dynamics equilibrium.
- **Network type**: Try different networks and decide which one fits best our research.
- **Heterogeneous β** : Give different social norm sensitivities to different network communities.

- Multi layer networks: Separate physical-proximity networks from information-sharing networks.
- Validation: Compare the model output to empirical data on adoption.

9 Limitations

The model we are using, given that it is a first approach, is a very simplistic model. Because of this, it has several limitations which could develop to potential issues.

- First, the no reversion decision. The state pipeline with only one way may be too optimistic for some sustainability scenarios.
- Also, the homogeneous β and γ , global constants that in reality may vary by household type, income and location.
- Furthermore, the absence of tipping points. The linear probability formula we are using does not capture threshold effects, like 'I will only adopt once 30% of my neighbours have'.
- The seed sensitivity limitation also has an important role on the limitations.
- Finally, the lack of ability to validate our result against real data. As it is now, the model is completely conceptual. Calibration to real adoption datasets would increase its predictive power

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